EVA: Smart College Event & Venue Management Platform

Software Requirements Specification

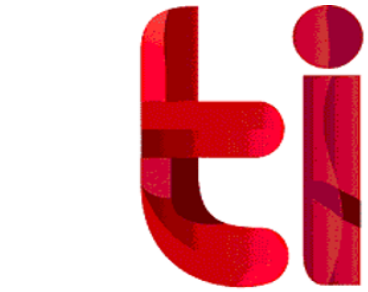
For Software Engineering (UCS 503P)

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List of Acronyms

Acronym	Meaning
SRS	Software Requirements Specification
FR	Functional Requirement
NFR	Non-Functional Requirement
UI	User Interface
API	Application Programming Interface
ER	Entity-Relationship
DB	Database
HTTP	Hypertext Transfer Protocol
HTTPS	Hypertext Transfer Protocol Secure
ID	Identifier
QR	Quick Response (code)
AI	Artificial Intelligence
MVP	Minimum Viable Product
USP	Unique Selling Proposition

Chapter 1: Introduction

1.1 Purpose

The purpose of this document is to present a detailed Software Requirements Specification (SRS) for EVA, a unified Smart College Event & Venue Management Platform. EVA connects event creation, venue booking, student registration, QR-based footfall tracking, feedback collection, and analytics so that college club coordinators can manage an event from planning through post-event evaluation in a single system.

Existing approaches to campus event management are fragmented: venue availability is checked separately, registrations are collected through disconnected forms, communication happens across different channels, and feedback and footfall are recorded and analyzed manually. EVA is intended to replace this fragmented workflow with one connected platform.

This document is intended to be used by the development team, the course instructor/evaluator, and any future contributor to the project as the single source of truth for what the system is required to do.

The system provides, among others, the following headline capabilities:

- Centralized creation and management of college club events
- Venue search, availability checking, and conflict-free booking
- AI-assisted smart venue recommendation based on event requirements
- Student-facing event discovery and registration
- QR-code-based footfall tracking to capture actual attendance
- Structured post-event feedback collection
- Analytics connecting registrations, footfall, and feedback for event performance insight

1.2 Document Conventions

Functional requirements are labelled FR-01 through FR-15 and are grouped by feature area (Chapter 3). Non-functional requirements are grouped by quality attribute in Chapter 5. Requirements written with *shall* are mandatory; requirements written with *should/may* are desirable but not mandatory for the first release.

1.3 Intended Audience and Reading Suggestions

This document is intended for:

- The development team, to understand what must be built.
- The course instructor/evaluator, to assess completeness and correctness of the requirements.
- Future maintainers, who may extend the system beyond the initial release.

Readers interested only in what the system does should read Chapters 2–4; readers interested in quality attributes and technical design should also read Chapters 5–8.

1.4 Project Scope

EVA is a web-based platform scoped specifically to a single college's clubs, venues, students, and events for the initial release. A full discussion of in-scope and out-of-scope items is given in Section 2.4.

1.5 Definitions, Acronyms and Abbreviations

See the List of Acronyms on page 5 for abbreviations used throughout this document. Key domain terms are defined here:

- Event — An activity created by a Club Coordinator that students can discover, register for, and attend.
- Venue — A bookable campus space with a defined capacity and set of facilities.
- Booking — A confirmed assignment of a venue to an event for a specific date and time.
- Footfall — The number of registered students who actually attended an event, captured via QR check-in.
- Club Coordinator — A student or staff member authorized to create events on behalf of a college club.

1.6 Document Overview

The remainder of this document is organized as follows: Chapter 2 gives the overall description of the product, its stakeholders, and its scope. Chapter 3 lists all functional requirements grouped by module. Chapter 4 describes external interface requirements. Chapter 5 lists non-functional requirements. Chapter 6 covers system architecture and technology choices. Chapter 7 covers the database design. Chapter 8 presents the use-case model. Chapter 9 walks through an end-to-end example scenario. Chapters 10–12 cover

key differentiators, future enhancements, and the conclusion. The appendices contain a requirements traceability matrix and a glossary.

Chapter 2: Overall Description

2.1 Product Perspective

EVA is a new, self-contained, web-based platform designed specifically for college club event and venue management. It is not a component of an existing system; it acts as a connecting layer between club coordinators who need to plan events and book venues, and students who wish to discover, attend, and give feedback on those events. Figure 2.1 shows the basic end-to-end workflow of the system.

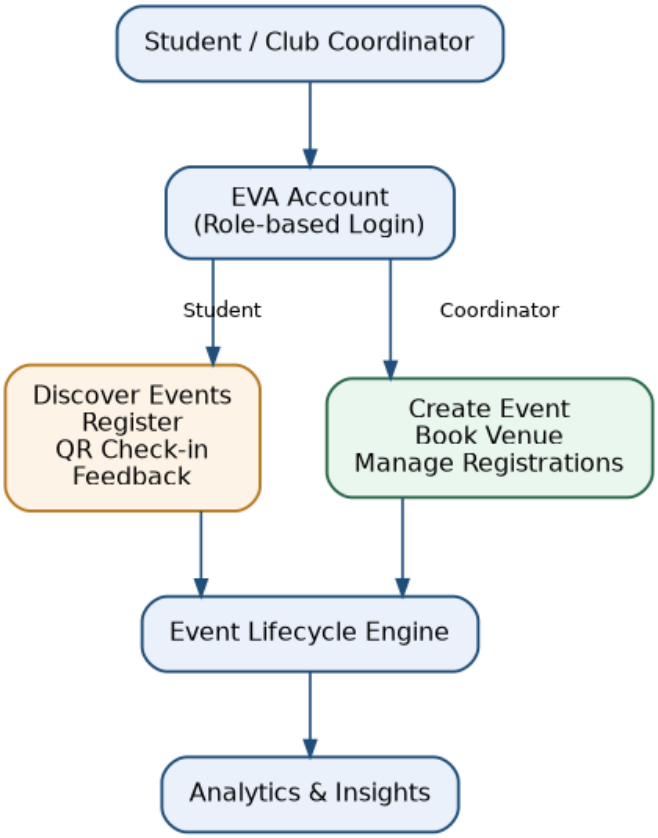


Figure 2.1: Basic end-to-end workflow of EVA

2.2 Problem Statement

College clubs often use multiple disconnected tools to organize an event. Venue availability may be checked separately, registrations may be collected through forms, communication may happen across different channels, footfall may be recorded independently, and feedback and analysis may be performed manually.

This fragmentation creates several practical problems:

- Manual venue booking and the possibility of scheduling conflicts.
- Scattered registration information and additional coordination effort.
- A gap between the number of registrations and actual event footfall.
- Feedback and event-performance information remaining disconnected.
- Limited consolidated data for improving future events.

The central problem addressed by EVA is therefore the lack of a single, college-focused platform that connects the complete event lifecycle.

2.3 Objectives

The main objectives of EVA are to:

- Provide a centralized platform for college club events.
- Simplify venue selection and booking while checking availability, capacity, and conflicts.
- Allow students to discover and register for college events through one platform.
- Record actual participation using QR-based footfall tracking.
- Collect structured feedback after events.
- Provide analytics based on registrations, footfall, feedback, and event performance.
- Introduce AI-powered smart venue recommendation as part of the MVP.

2.4 Scope of the Project

2.4.1 In Scope

The first release of the system will include:

- Role-based registration/login for Students and Club Coordinators
- Event creation and management
- Venue directory, filtering, availability checking, and conflict detection
- Event discovery, search, and filtering for students
- Event registration with capacity tracking
- QR-code generation and QR-based check-in
- Post-event feedback and ratings
- Analytics dashboard for coordinators (registrations, footfall, feedback)

2.4.2 Out of Scope

For the initial version, the following are not included:

- Online payment or ticketing gateway integration
- Support for multiple colleges/campuses
- A dedicated platform administrator role and institution-wide venue master-data management (venues are assumed to be pre-populated for the MVP)
- Native mobile applications (the MVP is a responsive web application)
- Integration with institutional calendars or external logistics/catering vendors
- Advanced, self-learning recommendation models beyond a constrained, explainable AI recommendation layer

2.5 Stakeholders

Table 2.1: Project stakeholders

Stakeholder	Role
Students	Primary users who discover, register for, attend, and give feedback on events
Club Coordinators	Students/staff who create events, book venues, and review analytics
College Clubs	Organizations on whose behalf events are created and managed
Institute Administration	Owns the campus venues used by the platform (venue data provider)
Development Team	Designs, develops, tests, and maintains the system

2.6 User Classes and Characteristics

2.6.1 Student

A member of the college community who discovers events, registers for them, checks in via QR code at the venue, and submits feedback after attending.

2.6.2 Club Coordinator

A student or staff member authorized to act on behalf of a college club. Coordinators create and manage events, search for and book suitable venues (assisted by AI recommendations), monitor registrations and footfall, and review analytics for their events.

A single person may act only as a Student for personal participation and separately as a Club Coordinator for club responsibilities; the platform supports both roles being associated with the same underlying account where applicable.

2.7 Operating Environment

EVA is a web application accessible from any modern desktop or mobile browser with an active internet connection. The backend is expected to run on cloud/server infrastructure reachable over Hypertext Transfer Protocol Secure (HTTPS). Detailed technology choices are discussed in Chapter 6.

2.8 Design and Implementation Constraints

- Only users with valid institute credentials can register as Students or Club Coordinators.
- Internet connectivity is required to use the platform.
- The master list of campus venues (capacity, facilities, location) is assumed to be seeded/maintained outside the two primary user roles for the MVP; institution-level venue administration is a future enhancement (see Chapter 11).
- QR check-in requires the student's device (or a printed code) to be scannable at the venue entry point.
- The AI venue recommendation layer must be deterministic and explainable in the MVP; it is introduced only after the core booking workflow is functional.
- The initial version does not process online payments or ticket sales.

2.9 Assumptions and Dependencies

- Students and coordinators have access to a valid institute account/ID.
- Coordinators provide accurate event requirements (capacity, timing, facilities).
- Venue capacity and facility data made available to the system is accurate and kept reasonably up to date.
- Users have internet access and a device capable of displaying or scanning a QR code.
- Historical utilization data accumulates over time to improve the quality of AI venue recommendations.

Chapter 3: Functional Requirements

This chapter lists all functional requirements of EVA, grouped by feature module. Each requirement is uniquely identified (FR-XX) so that it can be traced through design, implementation, and testing (see Appendix A).

3.1 Authentication and Profile

FR-01: User Registration The system shall allow a person to create an account as a Student or as a Club Coordinator using institute credentials.

- User selects a role: Student or Club Coordinator.
- User enters institute email/ID.
- System validates the credentials.
- User creates a password and completes profile details (for coordinators, the associated club).
- Account becomes active only after successful validation.

FR-02: User Login The system shall allow registered users to securely log in using their institute email/ID and password. The system shall authenticate the credentials and route the user to the appropriate role-based dashboard before granting access.

FR-03: User Profile Users shall be able to:

- View and edit their profile information.
- (Coordinators) view their club affiliation and the events they manage.
- (Students) view events they have registered for and their feedback history.

3.2 Event Management

FR-04: Create Event A Club Coordinator shall be able to create an event. An event may contain the fields shown in Table 3.1.

Table 3.1: Fields captured when creating an event

Field	Description
Event title	Short name of the event
Club	Organizing club
Description	Free-text description of the event
Category	e.g. Technical, Cultural, Sports, Workshop
Date & time	Scheduled start and end time
Expected footfall	Anticipated number of attendees
Required facilities	e.g. projector, sound system, seating layout
Status	Draft / Published / Completed / Cancelled

Example event: "AI Hackathon Night", category Technical, expected footfall 120, required facilities projector and sound system, organized by the Coding Club.

FR-05: Edit/Cancel Event The Club Coordinator shall be able to:

- Edit event information before it is published.
- Update event status (e.g. mark as completed).
- Cancel an event, which shall release the associated venue booking and notify registered students.

3.3 Venue and Booking

FR-06: Venue Directory and Filtering The system shall allow a Club Coordinator to search and filter campus venues by capacity, facilities, and location while booking a venue for an event, as shown in Table 3.2.

Table 3.2: Fields captured when booking a venue

Field	Description
Venue name	Name/identifier of the campus venue
Capacity	Maximum number of attendees supported
Facilities	e.g. projector, sound system, air-conditioning
Location	Building/block on campus
Availability window	Requested date and time slot for the booking

FR-07: Availability Check and Conflict Detection The system shall check whether a candidate venue is free for the requested date/time and shall prevent a booking that conflicts with an existing confirmed booking for the same venue and time slot.

FR-08: AI-Assisted Venue Recommendation The system shall rank suitable, available venues using event requirements such as capacity, timing, required facilities, and historical utilization, and present the ranked list to the Club Coordinator as recommendations. Figure 3.1 shows this workflow.

FR-09: Booking Confirmation The Club Coordinator shall be able to confirm a booking from the list of suitable/recommended venues, after which the venue shall be marked unavailable for that date/time slot.

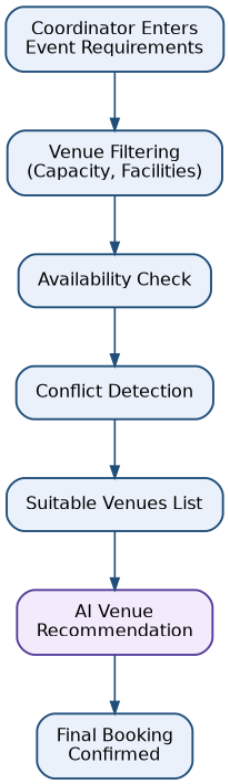


Figure 3.1: Event and Venue Booking workflow

3.4 Student Registration and Discovery

FR-10: Event Discovery Students shall be able to browse, search, and filter published events by club, category, date, and venue.

FR-11: Event Registration Students shall be able to register for a published event. The system shall track registration counts against the venue capacity and shall prevent registration once an event reaches full capacity.

Figure 3.2 shows the student-side workflow, from discovering an event through to submitting feedback after attending.

3.5 QR Footfall

FR-12: QR Code Generation The system shall generate a unique QR code for each confirmed event registration.

FR-13: QR Check-in The system shall allow a coordinator (or a check-in device/application) to scan a student's QR code at the venue entrance and shall record a timestamped check-in, distinguishing actual attendance (footfall) from registration.

3.6 Feedback

FR-14: Post-Event Feedback After an event is marked completed, students who checked in shall be able to submit a rating (e.g. five-star scale) and a free-text comment about the event and venue.

3.7 Analytics

FR-15: Event and Venue Analytics Dashboard The system shall provide Club Coordinators with an analytics dashboard connecting registrations, footfall, and feedback for their events, including turnout rate (footfall vs. registrations) and venue utilization history, which shall also feed the AI venue recommendation (FR-08).

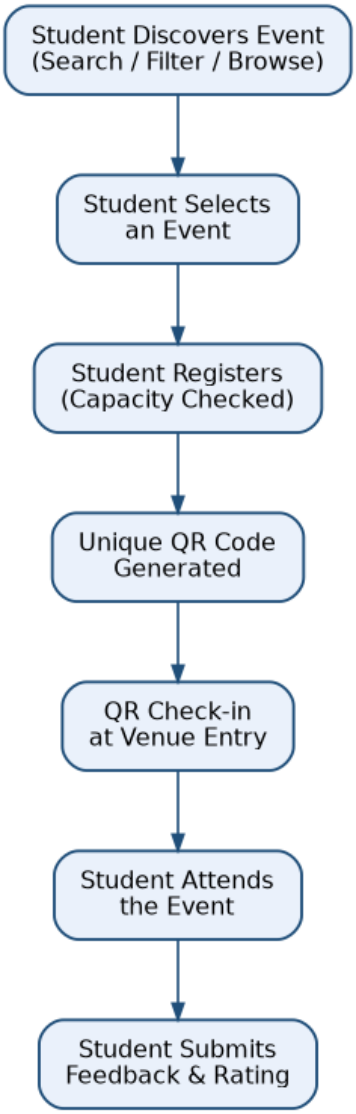


Figure 3.2: Student Event Discovery and Registration workflow

Chapter 4: External Interface Requirements

4.1 User Interfaces

The system shall expose a responsive web interface usable on both desktop and mobile browsers, with the following principal screens: registration/login, role-based dashboard, event discovery/browse, event detail, create/edit event, venue booking and recommendations, QR check-in, feedback, and analytics.

4.2 Hardware Interfaces

EVA does not require any dedicated or proprietary hardware. It communicates only with standard hardware available on client devices (camera, for scanning/displaying QR codes; touchscreen/keyboard/mouse, for input).

4.3 Software Interfaces

- Web browser: any modern browser (Chrome, Firefox, Edge, Safari).
- Database: a relational or document database for persisting users, clubs, venues, events, registrations, check-ins, and feedback (see Chapter 7).
- QR code generation/scanning library or service.

4.4 Communication Interfaces

All client–server communication shall occur over HTTPS.

Chapter 5: Non-Functional Requirements

5.1 Security

The system shall:

- Restrict registration to users with valid institute credentials.
- Store passwords securely (e.g. salted and hashed, never in plain text).
- Use authentication and authorization for all protected actions, including role-based access for Student vs. Club Coordinator features.
- Ensure QR codes are unique per registration and cannot be trivially reused for a different student's check-in.
- Protect user and event data in line with the platform's privacy policy.

5.2 Performance

The system should:

- Load major pages within a reasonable time under normal load.
- Return venue availability and recommendation results quickly.
- Support multiple concurrent users, including bursts of registrations for popular events.
- Process QR check-ins with minimal delay at venue entry points.

5.3 Reliability

The system should:

- Maintain consistent booking data and prevent double-booking of a venue.
- Prevent accidental loss of registration, footfall, and feedback data.
- Handle server failures gracefully.
- Maintain an accurate, append-only check-in record for footfall reporting.

5.4 Usability

The application should provide simple navigation, a mobile-friendly design, clear event categories, an easy event-creation and venue-booking flow, straightforward event discovery and registration for students, and a simple QR check-in and feedback experience.

5.5 Scalability

The architecture should allow the platform to support increasing numbers of clubs, venues, events, students, and registrations without a redesign, and should be extensible to multiple campuses or institutions in future versions while preserving the same role-based workflow.

5.6 Maintainability

The system should use a modular architecture — separating event management, venue booking, registration, QR footfall, feedback, and analytics — so that features can be added or modified (for example, richer recommendation models or additional administrative roles) without affecting the rest of the application.

Chapter 6: System Architecture and Technology

6.1 Hardware Requirements

Minimum Client Requirements

A smartphone, laptop, or desktop with an internet connection, a modern web browser, and (for QR check-in) a camera.

Server Requirements

Depending on the chosen deployment: cloud/server infrastructure, database storage, an application server, and a service for generating/validating QR codes.

6.2 Software Requirements / Suggested Technology Stack

Table 6.1 shows one possible technology stack. This stack is a suggestion only; the SRS specifies what the system must do, and the final technology choice may be finalized separately by the team.

Table 6.1: Suggested technology stack

Component	Suggested Technology
Frontend	React.js / HTML, CSS, JavaScript
Backend	Node.js + Express
Database	MongoDB / PostgreSQL
Authentication	Institute email/ID verification
QR Code	QR generation and scanning library (e.g. qrcode, zxing)
Hosting	Vercel / Render / AWS
Version Control	Git + GitHub

6.3 System Architecture

Figure 6.1 shows a high-level architecture in which the client communicates with a backend API, which in turn talks to the database, a QR check-in service, and an AI recommendation engine.

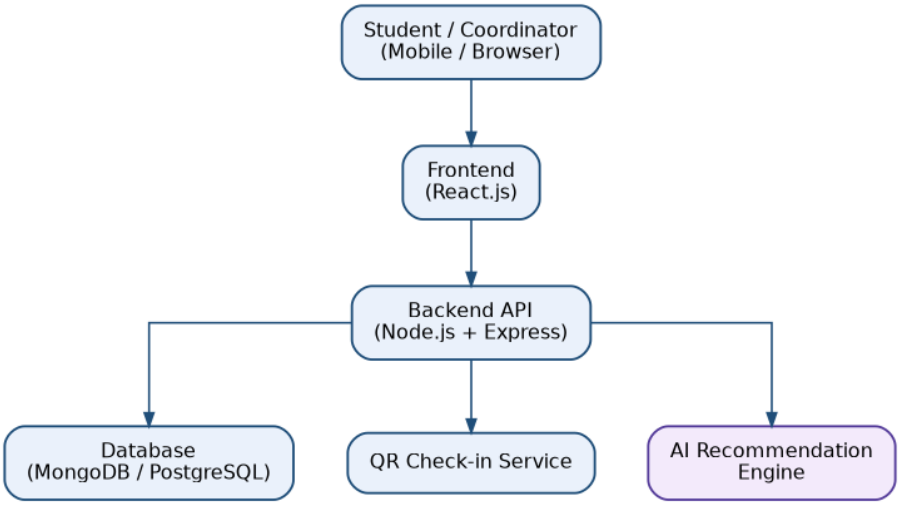


Figure 6.1: High-level system architecture

Chapter 7: Database Design

This chapter describes the major data entities of EVA and their attributes. A full physical schema (data types, indices, constraints) can be derived from this logical model during implementation.

7.1 User

Table 7.1: User entity

Attribute	Description
User_ID	Primary key
Name	User's full name
Email	Institute email address, used for verification
Password	Hashed password
Role	Student or Club Coordinator
Club_ID	Foreign key to Club (applicable for Coordinators)
Created_Date	Timestamp of account creation

7.2 Club

Table 7.2: Club entity

Attribute	Description
Club_ID	Primary key
Name	Club name
Description	Free-text description of the club

7.3 Venue

Table 7.3: Venue entity

Attribute	Description
Venue_ID	Primary key
Name	Venue name/identifier
Capacity	Maximum number of attendees supported
Facilities	List of available facilities (e.g. projector, sound system)
Location	Building/block on campus
Status	Active / under maintenance

7.4 Event

Table 7.4: Event entity

Attribute	Description
Event_ID	Primary key
Club_ID	Foreign key to Club
Coordinator_ID	Foreign key to User (creator)
Title	Event title
Description	Free-text description
Category	Event category
Date_Time	Scheduled start and end time
Venue_ID	Foreign key to Venue (once booked)
Expected_Footfall	Anticipated number of attendees
Status	Draft / Published / Completed / Cancelled

7.5 Registration

Table 7.5: Registration entity

Attribute	Description
Registration_ID	Primary key
Event_ID	Foreign key to Event
Student_ID	Foreign key to User (student)
QR_Code	Unique QR code issued for this registration
Registration_Date	Timestamp of registration
Status	Registered / Cancelled

7.6 CheckIn (Footfall)

Table 7.6: CheckIn (Footfall) entity

Attribute	Description
CheckIn_ID	Primary key
Registration_ID	Foreign key to Registration
Scan_Timestamp	Time the QR code was scanned at the venue

7.7 Feedback

Table 7.7: Feedback entity

Attribute	Description
Feedback_ID	Primary key
Event_ID	Foreign key to Event
Student_ID	Foreign key to User (reviewer)
Rating	Numeric rating (e.g. 1–5)
Comment	Free-text comment
Date	Date of feedback submission

7.8 Entity Relationships

A Club has many Coordinators (Users) and many Events. A Venue may be booked by many Events over time, but not by two Events for an overlapping date/time. An Event has many Registrations from Students, and a Registration has at most one CheckIn. After an Event is completed, Students who checked in may leave one Feedback entry each. His-

torical Event, Venue, and Feedback data together inform the AI venue recommendation used during booking.

Chapter 8: Use Case Model

8.1 Actors

EVA has two principal actors for the MVP: Student (who discovers, registers for, attends, and reviews events) and Club Coordinator (who creates events, books venues, and reviews analytics on behalf of a club).

8.2 Use Case Diagram

Figure 8.1 summarizes the major use cases available to each actor.

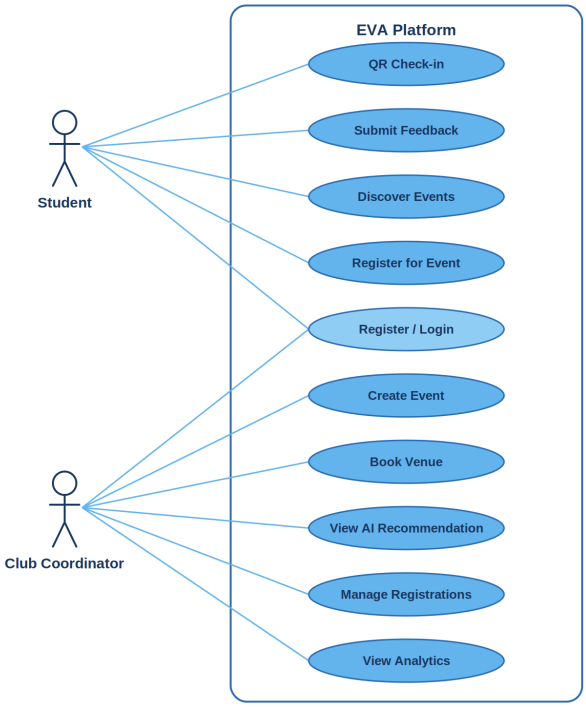


Figure 8.1: Use case diagram for EVA

8.3 Use Case Descriptions

Table 8.1 summarizes the primary use cases; each maps to one or more functional requirements from Chapter 3.

Table 8.1: Use case summary

Use Case	Description	Related FR
Register / Login	User creates an account and signs in using institute credentials	FR-01, FR-02
Create Event	Coordinator defines a new event and its requirements	FR-04, FR-05
Book Venue	Coordinator searches, filters, and confirms a venue booking	FR-06, FR-07, FR-09
View AI Recommendation	Coordinator reviews ranked, suitable available venues	FR-08
Discover Events	Student browses, searches, and filters events	FR-10
Register for Event	Student registers for a published event	FR-11, FR-12
QR Check-in	Student's attendance is captured via QR scan at the venue	FR-13
Submit Feedback	Student rates and comments on an attended event	FR-14
Manage Registrations	Coordinator monitors registrations and footfall for an event	FR-11, FR-13
View Analytics	Coordinator reviews turnout, feedback, and venue utilization	FR-15

Chapter 9: Example User Scenario

Consider a Club Coordinator, Rohan, from the Coding Club who wants to organize a hackathon night, and a student, Riya, who wishes to attend. Figure 9.1 walks through the end-to-end interaction with EVA, which exercises event creation, AI-assisted venue booking, registration, QR check-in, feedback, and analytics in a single workflow.

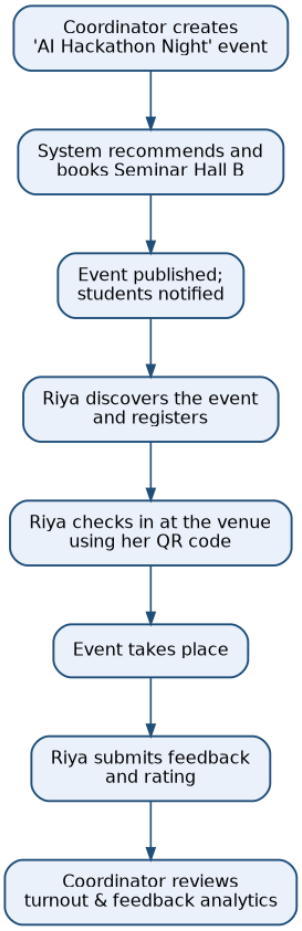


Figure 9.1: End-to-end example scenario

Chapter 10: Key Features and Unique Selling Points

Three differentiators make EVA stand out from generic event or campus platforms.

10.1 Smart Venue Recommendation

AI-assisted ranking of suitable, available venues using event requirements such as capacity, timing, facilities, and historical utilization. Problem solved: manual venue matching and avoidable scheduling conflicts.

10.2 Connected Event Intelligence

A connected flow from Registration → Footfall → Feedback → Analytics, enabling a more complete view of event performance than registration counts alone. Problem solved: disconnected event-performance data.

10.3 College-Club Focus

A workflow designed specifically around Students, Clubs, Campus Venues, and College Events, rather than a general-purpose public-event orientation. Problem solved: generic marketplaces and event tools are not optimized for the specific operational needs of college clubs.

Chapter 11: Future Enhancements

Future versions of EVA could include:

- A dedicated platform/institution-administrator role for venue master-data management
- Support for multiple clubs, departments, campuses, or institutions
- Richer analytics and reporting
- Integration with institutional calendars
- More advanced, self-learning recommendation models
- Native Android/iOS applications
- Online payment/ticketing for paid events
- Campus map integration for venue navigation

Chapter 12: Conclusion

EVA aims to provide a secure, convenient, and college-club-focused platform for event and venue management. By combining event creation, AI-assisted venue booking, student registration, QR-based footfall tracking, structured feedback, and connected analytics, the platform addresses the fragmentation present in current campus event workflows.

The project is deliberately scoped around two user roles — Students and Club Coordinators — and five core modules, making it practical for incremental development within

a college Software Engineering project. The system is intended to make campus events easier to plan, more measurable, and more useful for improving future events, following the cycle Plan → Organize → Measure → Improve.

Chapter A: Requirements Traceability Matrix

Table A.1 maps each functional requirement to the feature module and the use case(s) that exercise it, to support later test planning.

Table A.1: Requirements traceability matrix

ID	Requirement	Module	Use Case
FR-01	User Registration	Authentication	Register / Login
FR-02	User Login	Authentication	Register / Login
FR-03	User Profile	Authentication	Register / Login
FR-04	Create Event	Event Management	Create Event
FR-05	Edit/Cancel Event	Event Management	Create Event
FR-06	Venue Directory and Filtering	Venue & Booking	Book Venue
FR-07	Availability Check and Conflict Detection	Venue & Booking	Book Venue
FR-08	AI-Assisted Venue Recommendation	Venue & Booking	View AI Recommendation
FR-09	Booking Confirmation	Venue & Booking	Book Venue
FR-10	Event Discovery	Student Registration	Discover Events
FR-11	Event Registration	Student Registration	Register for Event
FR-12	QR Code Generation	QR Footfall	Register for Event
FR-13	QR Check-in	QR Footfall	QR Check-in
FR-14	Post-Event Feedback	Feedback	Submit Feedback
FR-15	Event and Venue Analytics Dashboard	Analytics	View Analytics

Chapter B: Glossary

- Student — A member of the college community who discovers, registers for, attends, and reviews events.
- Club Coordinator — A student or staff member who creates events and books venues on behalf of a college club.
- Event — An activity created by a Club Coordinator that students can discover and register for.
- Venue — A bookable campus space with a defined capacity and set of facilities.
- Booking — A confirmed assignment of a venue to an event for a specific date and time.
- Footfall — The number of registered students who actually attended an event, captured via QR check-in.
- QR Check-in — The process of scanning a student's unique QR code at a venue to record attendance.
- AI Venue Recommendation — A ranked list of suitable, available venues generated using event requirements and historical utilization data.